Data Models Query Language

Lecture 2: SQL

Week 3

Agenda

1. **Set operators & nested queries:** How to combine the results of multiple queries correctly and efficiently.
2. **Aggregation & GROUP BY:** How to compute summary statistics over your data.
3. **Advanced Topics: NULLs and Joins:** Handling missing data and understanding different join semantics.

1. Set Operators & Nested Queries

5

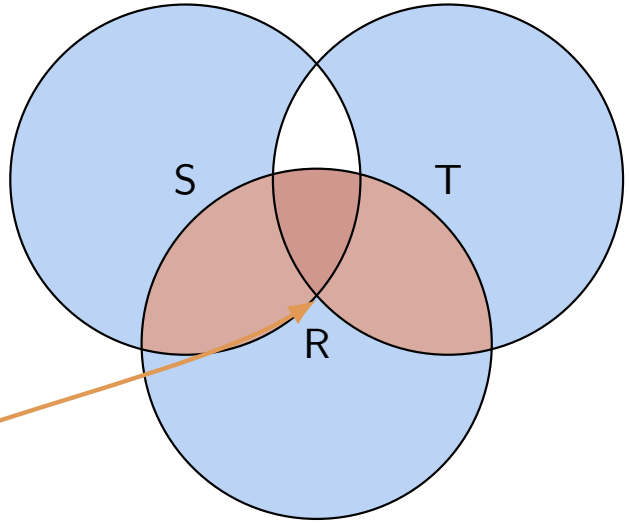
An Unintuitive Query

```
SELECT DISTINCT R.A
FROM R, S, T
WHERE R.A=S.A OR R.A=T.A
```

What does it compute?

An Unintuitive Query

```
SELECT DISTINCT R.A
FROM R, S, T
WHERE R.A=S.A OR R.A=T.A
```



Computes $R \cap (S \cup T)$

But what if $S = \phi$?

Go back to the semantics!

An Unintuitive Query

```
SELECT DISTINCT R.A
FROM R, S, T
WHERE R.A=S.A OR R.A=T.A
```

Recall the semantics!

1. Take cross-product
2. Apply selections / conditions
3. Apply projection

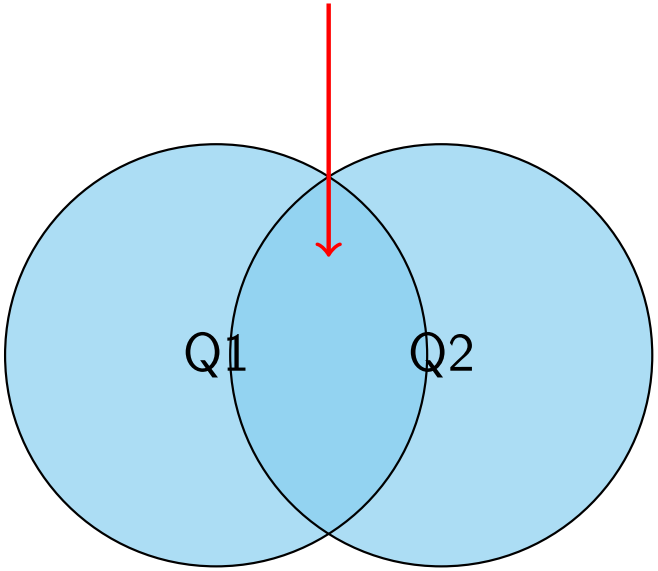
If $S = \{\}$, then the cross product of R, S, $T = \{\}$,
and the query result $= \{\}$!

Must consider semantics here.
Are there more explicit way to do set operations like this?

Explicit Set Operators: INTERSECT

```
SELECT R.A
FROM R, S
WHERE R.A=S.A
INTERSECT
SELECT R.A
FROM R, T
WHERE R.A=T.A
```

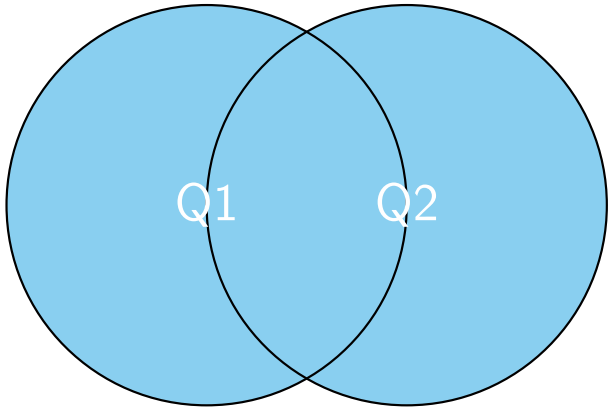
$$\{r.A \mid r.A = s.A\} \cap \{r.A \mid r.A = t.A\}$$



UNION

```
SELECT R.A
FROM R, S
WHERE R.A=S.A
UNION
SELECT R.A
FROM R, T
WHERE R.A=T.A
```

$$\{r.A \mid r.A = s.A\} \cup \{r.A \mid r.A = t.A\}$$



By default: SQL uses set semantics!

What if we want duplicates?

UNION ALL

```
SELECT R.A
FROM R, S
WHERE R.A=S.A
UNION ALL
SELECT R.A
FROM R, T
WHERE R.A=T.A
```

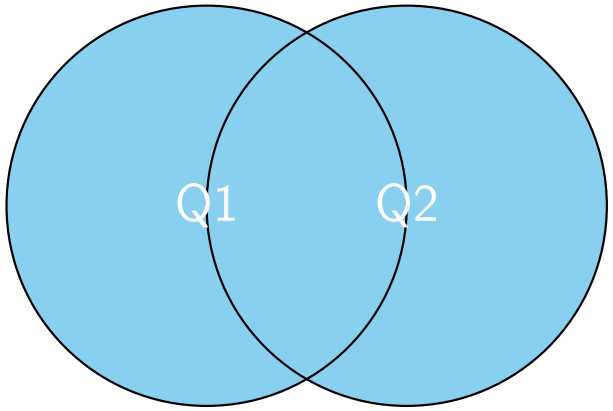
$$\{r.A \mid r.A = s.A\} \cup \{r.A \mid r.A = t.A\}$$

ALL indicates Multiset operations

UNION ALL

```
SELECT R.A
FROM R, S
WHERE R.A=S.A
UNION ALL
SELECT R.A
FROM R, T
WHERE R.A=T.A
```

$$\{r.A \mid r.A = s.A\} \cup \{r.A \mid r.A = t.A\}$$

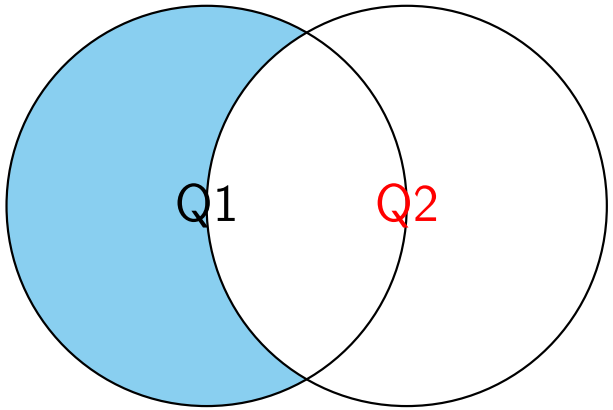


ALL indicates Multiset operations

UNION ALL

```
SELECT R.A
FROM R, S
WHERE R.A=S.A
EXCEPT
SELECT R.A
FROM R, T
WHERE R.A=T.A
```

$$\{r.A \mid r.A = s.A\} \setminus \{r.A \mid r.A = t.A\}$$



INTERSECT: Still some subtle problems...

Company(name, hq_city)
Product(pname, maker, factory_loc)

```
SELECT hq_city
FROM  Company, Product
WHERE maker = name
    AND factory_loc = 'US'
INTERSECT
SELECT hq_city
FROM  Company, Product
WHERE maker = name
    AND factory_loc = 'China'
```

"Headquarters of companies which make products in US AND China"

What can go wrong here?

INTERSECT: Remember the semantics!

```
Company(name, hq_city) AS C
Product(pname, maker,
factory_loc) AS P
```

Example: C JOIN P on maker = name

C.name	C.hq_city	P.pname	P.maker	P.factory_loc
X Co.	Seattle	X	X Co.	U.S.
Y Inc.	Seattle	X	Y Inc.	China

```
SELECT hq_city
FROM Company, Product
WHERE maker = name
  AND factory_loc='US'
INTERSECT
SELECT hq_city
FROM Company, Product
WHERE maker = name
  AND factory_loc='China'
```

X Co has a factory in the US (but not China)
Y Inc. has a factory in China (but not US)

But Seattle is returned by the query!

One Solution: Nested Queries

Company(name, hq_city)
Product(pname, maker, factory_loc)

"Headquarters of companies which make products in US AND China"

```
SELECT DISTINCT hq_city
FROM Company, Product
WHERE maker = name
   AND name IN (
       SELECT maker
       FROM Product
       WHERE factory_loc = 'US')
   AND name IN (
       SELECT maker
       FROM Product
       WHERE factory_loc = 'China')
```

High-level note on nested queries

We can do nested queries because SQL is **compositional**:

- Everything (inputs / outputs) is represented as multisets
- The output of one query can thus be used as the input to another (nesting)!

This is extremely powerful!

Nested queries: Sub-queries Return Relations

Example

Company(name, city)
Product(name, maker)
Purchase(id, product, buyer)

"Cities where one can find companies that manufacture products bought by Joe Blow"

```
SELECT c.city
FROM  Company c
WHERE  c.name  IN (
      SELECT pr.maker
  FROM  Purchase p, Product pr
  WHERE  p.product = pr.name
      AND p.buyer = 'Joe Blow')
```

Nested Queries

Are these queries equivalent?

```
SELECT c.city
FROM  Company c
WHERE  c.name  IN (
  SELECT pr.maker
  FROM   Purchase p, Product
    pr
  WHERE  p.name = pr.product
    AND p.buyer = 'Joe Blow')
```

```
SELECT c.city
FROM  Company c,
  Product pr,
  Purchase p
WHERE  c.name = pr.maker
 AND  pr.name = p.product
 AND  p.buyer = 'Joe Blow'
```

Beware of duplicates!

Nested Queries

```
SELECT DISTINCT c.city
FROM  Company c,
  Product pr,
  Purchase p
WHERE  c.name = pr.maker
AND  pr.name = p.product
AND  p.buyer = 'Joe Blow'
```

```
SELECT DISTINCT c.city
FROM  Company c
WHERE  c.name IN (
 SELECT pr.maker
 FROM  Purchase p, Product pr
 WHERE  p.product = pr.name
        AND p.buyer = 'Joe Blow
    ')
```

Now they are equivalent (both use set semantics)

Subqueries Return Relations

You can also use operations of the form:

- s > ALL R
- s < ANY R
- EXISTS R

ANY and ALL not supported by SQLite.

Ex: Product(name, price, category, maker)

```
SELECT name
FROM Product
WHERE price > ALL(
      SELECT price
   FROM Product
   WHERE maker = 'Gizmo-Works')
```

Find products that are more expensive than all those produced by "Gizmo-Works"

Subqueries Return Relations

You can also use operations of the form:

- s > ALL R
- s < ANY R
- EXISTS R

Ex: Product(name, price, category, maker)

```
SELECT p1.name
FROM Product p1
WHERE p1.maker ='Gizmo-Works'
  AND EXISTS(
       SELECT p2.name
   FROM Product p2
   WHERE p2.maker <> 'Gizmo-
    Works'
       AND p1.name = p2.name)
```

```
<> means !=
```

Find 'copycat products, i.e. products made by competitors with the same names as products made by Gizmo-Works"

Correlated Queries Using External Vars in Internal Subquery

Movie(title, year, director, length)

```
SELECT DISTINCT title
FROM   Movie AS m
WHERE  year < ANY(
           SELECT year
       FROM    Movie
       WHERE title = m.title)
```

Find movies whose title appears more than once by finding any version of a movie that has another version from a later year. It effectively returns the titles of the older versions in a series of remakes or re-releases.

Note the scoping of the variables!

Complex Correlated Query

Product(name, price, category, maker, year)

Find products (and their manufacturers) that are more expensive than all products made by the same manufacturer before 1972

```
SELECT DISTINCT x.name, x.maker
FROM Product AS x
WHERE x.price > ALL(
            SELECT y.price
      FROM Product AS y
      WHERE x.maker = y.maker
            AND y.year < 1972)
```

Can be very powerful (also much harder to optimize)

Correlated vs Regular Subqueries

In terms of execution

- Regular: executed once for the entire outer query
- Correlated: executed once for each row processed by the outer query (due to the dependence between inner and outer queries)

This means that correlated subqueries are usually very slow

- When possible, rewrite using JOINs for better performance

```
SELECT DISTINCT title
FROM Movie AS m
WHERE year <> ANY(
      SELECT year
      FROM Movie
      WHERE title = m.title)
```

Correlated vs Regular Subqueries

In terms of execution

- Regular: executed once for the entire outer query
- Correlated: executed once for each row processed by the outer query (due to the dependence between inner and outer queries)

This means that correlated subqueries are usually very slow

- When possible, rewrite using JOINs for better performance

```
SELECT DISTINCT title
FROM Movie AS m
WHERE year <> ANY(
      SELECT year
      FROM Movie
      WHERE title = m.title)
```

```
SELECT DISTINCT m1.title
FROM Movie m1 JOIN Movie m2
        ON m1.title=m2.title
WHERE m1.year<>m2.year
```

Execution Logic: Why Join > Correlated?

- **Correlated Subquery** ($O(N \times M)$):
  - Logic: "For every row in the outer table, run the inner query once."
  - If you have 1 million rows, the subquery runs 1 million times!
- **Uncorrelated (Regular) Subquery**:
  - Logic: Run the inner query *once*, store the result in memory, then filter the outer table.
- **JOIN**:
  - The Query Optimizer can use **Hash Joins** or **Merge Joins** to process everything in one pass ($O(N + M)$ or $O(N \log N)$).

Always default to a `JOIN` unless the query logic specifically requires existential quantification (`EXISTS`) that short-circuits faster than a join.

Basic SQL Summary

- SQL provides a high-level declarative language for manipulating data (DML)
- The workhorse is the SFW block
- Set operators are powerful but have some subtleties
- Powerful, nested queries also allowed.

2. Aggregation & GROUP BY

27

Aggregation

```
SELECT AVG(price)
FROM Product
WHERE maker = ``Toyota''
```

```
SELECT COUNT(*)
FROM Product
WHERE year > 1995
```

- SQL supports several **aggregation** operations:
  - SUM, COUNT, MIN, MAX, AVG
    - Except COUNT, all aggregations apply to a single attribute

Aggregation: COUNT

COUNT applies to duplicates, unless otherwise stated

```
SELECT COUNT(category)
FROM Product
WHERE year > 1995
```

Note: Same as COUNT(*). Why?

We probably want:

```
SELECT COUNT(DISTINCT category)
FROM Product
WHERE year > 1995
```

More Examples

Purchase(product, date, price, quantity)

```
SELECT SUM(price * quantity)
FROM Purchase
```

```
SELECT SUM(price * quantity)
FROM Purchase
WHERE product = 'bagel'
```

What do these mean?

Simple Aggregations

Purchase

Product	Date	Price	Quantity
bagel	10/21	1	20
banana	10/3	0.5	10
banana	10/10	1	10
bagel	10/25	1.50	20

```
SELECT SUM(price * quantity)
FROM Purchase
WHERE product = 'bagel'
```

$\Longrightarrow$ 50 (= 1*20 + 1.50*20)

Grouping and Aggregation

Purchase(product, date, price, quantity)

```
SELECT product,
       SUM(price * quantity) AS TotalSales
FROM Purchase
WHERE date > '10/1/2005'
GROUP BY product
```

Find total sales after 10/1/2005 per product.

Let's see what this means...

Grouping and Aggregation

Semantics of the query:

1. Compute the FROM and WHERE clauses
2. Group by the attributes in the GROUP BY
3. Compute the SELECT clause: grouped attributes and aggregates

1. Compute the FROM and WHERE clauses

```
SELECT product, SUM(price*quantity) AS TotalSales
FROM Purchase
WHERE date > '10/1/2005'
GROUP BY product
```

From $\Rightarrow$

Product	Date	Price	Quantity
Bagel	10/21	1	20
Bagel	10/25	1.50	20
Banana	10/3	0.5	10
Banana	10/10	1	10

2. Group by the attributes in the GROUP BY

```
SELECT product, SUM(price*quantity) AS TotalSales
FROM Purchase
WHERE date > '10/1/2005'
GROUP BY product
```

Product	Date	Price	Quantity
Bagel	10/21	1	20
Bagel	10/25	1.50	20
Banana	10/3	0.5	10
Banana	10/10	1	10

GROUP BY

$\Rightarrow$

Product	Date	Price	Quantity
Bagel	10/21	1	20
Bagel	10/25	1.50	20
Banana	10/3	0.5	10
Banana	10/10	1	10

3. Compute the SELECT clause: grouped attributes and aggregates

```
SELECT product, SUM(price*quantity) AS TotalSales
FROM Purchase
WHERE date > '10/1/2005'
GROUP BY product
```

Product	Date	Price	Quantity
Bagel	10/21	1	20
	10/25	1.50	20
Banana	10/3	0.5	10
	10/10	1	10

SELECT

$\Rightarrow$

Product	TotalSales
Bagel	50
Banana	15

HAVING Clause

```
SELECT product, SUM(price*quantity)
FROM Purchase
WHERE date > '10/1/2005'
GROUP BY product
HAVING SUM(quantity) > 100
```

Same query as before, except that we consider only products that have more than 100 buyers

HAVING clauses contains conditions on aggregates

Whereas WHERE clauses condition on individual tuples...

General form of Grouping and Aggregation

SELECT S
FROM $R_1, \ldots, R_n$
WHERE C_1
GROUP BY $a_1, \ldots, a_k$
HAVING C_2

- S = Can ONLY contain attributes $a_1, \ldots, a_k$ and/or aggregates over other attributes
- C_1 = is any condition on the attributes in $R_1, \ldots, R_n$
- C_2 = is any condition on the aggregate expressions

General form of Grouping and Aggregation

SELECT S
FROM $R_1, ..., R_n$
WHERE C_1
GROUP BY $a_1, ..., a_k$
HAVING C2

Evaluation steps:

1. Evaluate FROM-WHERE: apply condition C1 on the attributes in $R_1, ..., R_n$
2. GROUP BY the attributes $a_1, ..., a_k$
3. Apply condition C_2 to each group (may have aggregates)
4. Compute aggregates in S and return the result

Group-by v.s. Nested Query

Author(<u>login</u>, name)
Wrote(login, url)

Find authors who wrote > 10 documents:

- Attempt 1: with nested queries

```
SELECT DISTINCT Author.name
FROM Author
WHERE COUNT(
  SELECT Wrote.url
  FROM Wrote
  WHERE Author.login = Wrote.login) > 10
```

This is SQL by a novice

Group-by v.s. Nested Query

Find all authors who wrote at least 10 documents:

- Attempt 2: SQL style (with GROUP BY)

```
SELECT Author.name
FROM Author, Wrote
WHERE Author.login = Wrote.login
GROUP BY Author.name
HAVING COUNT(Wrote.url) > 10
```

This is SQL by an expert

No need for DISTINCT: automatically from GROUP BY

Group-by vs. Nested Query

Which way is more efficient?

- Attempt #1 - With nested: How many times do we do a SFW query over all of the Wrote relations?
- Attempt #2 - With group-by: How about when written this way?

With GROUP BY can be much more efficient!

3. Advanced SQL-izing

NULLS in SQL

- Whenever we don't have a value, we can put a NULL
- Can mean many things:
  - Value does not exists
  - Value exists but is unknown
  - Value not applicable
  - Etc.
- The schema specifies for each attribute if can be null (nullable attribute) or not
- How does SQL cope with tables that have NULLs?

Null Values

- For numerical operations, NULL $->$ NULL:
  - If $x =$ NULL then $4^*(3 - x)/7$ is still NULL
- For boolean operations, in SQL there are three values:

FALSE	$= 0$
UNKNOWN	$= 0.5$
TRUE	$= 1$

- If $x =$ NULL then $x =$ "*Joe*" is UNKNOWN

Null Values

- C1 AND C2 = min(C1, C2)
- C1 OR C2 = max(C1, C2)
- NOT C1 = 1 - C1

```
SELECT *
FROM Person
WHERE (age < 25)
AND (height > 6 AND weight > 190)
```

Won't return e.g. (age=20, height=NULL, weight=200)!

Rule in SQL: include only tuples that yield TRUE (1.0)

Null Values

Unexpected behavior:

```
SELECT *
FROM Person
WHERE age < 25 OR age >= 25
```

Does this query include all rows in the table?

Null Values

Can test for NULL explicitly:

- x IS NULL
- x IS NOT NULL

```
SELECT *
FROM Person
WHERE age < 25 OR age >= 25
OR age IS NULL
```

Now it includes all Persons!

Inner Joins

By default, joins in SQL are "inner joins":

Product(name, category)
Purchase(prodName, store)

```
SELECT Product.name, Purchase.store
FROM Product
JOIN Purchase ON Product.name = Purchase.prodName
```

```
SELECT Product.name, Purchase.store
FROM Product, Purchase
WHERE Product.name = Purchase.prodName
```

Both equivalent: Both INNER JOINS!

Inner Joins + NULLS = Lost data?

By default, joins in SQL are "inner joins":
Product(name, category)
Purchase(prodName, store)

```
SELECT Product.name, Purchase.store
FROM Product
JOIN Purchase ON Product.name = Purchase.prodName
```

```
SELECT Product.name, Purchase.store
FROM Product, Purchase
WHERE Product.name = Purchase.prodName
```

However: Products that never sold (with no Purchase tuple) will be lost!

Outer Joins

- An **outer join** returns tuples from the joined relations that don't have a corresponding tuple in the other relations

- I.e. If we join relations A and B on $a.X = b.X$, and there is an entry in A with $X = 5$, but none in B with $X = 5$.
  - A LEFT OUTER JOIN will return a tuple (a, NULL)!

Left outer joins in SQL:

```
SELECT Product.name, Purchase.store
FROM Product
  LEFT OUTER JOIN Purchase ON
    Product.name = Purchase.prodName
```

Now we'll get products even if they didn't sell

INNER JOIN:

Product

name	category
Gizmo	gadget
Camera	Photo
OneClick	Photo

Purchase

prodName	store
Gizmo	Wiz
Camera	Ritz
Camera	Wiz

```
SELECT Product.name, Purchase.store
FROM Product
INNER JOIN Purchase
ON Product.name = Purchase.prodName
```

$\Rightarrow$

name	store
Gizmo	Wiz
Camera	Ritz
Camera	Wiz

Note: another equivalent way to write an INNER JOIN!

LEFT OUTER JOIN:

Product

name	category
Gizmo	gadget
Camera	Photo
OneClick	Photo

Purchase

prodName	store
Gizmo	Wiz
Camera	Ritz
Camera	Wiz

```
SELECT Product.name, Purchase.store
FROM Product
LEFT OUTER JOIN Purchase
ON Product.name = Purchase.prodName
```

$\Rightarrow$

name	store
Gizmo	Wiz
Camera	Ritz
Camera	Wiz
OneClick	NULL

Other Outer Joins

- Left outer join:
  - Include the left tuple even if there's no match
- Right outer join:
  - Include the right tuple even if there's no match
- Full outer join:
  - Include the both left and right tuples even if there's no match